

$C(^{36}\text{Mg}, ^{35}\text{Mg}\gamma), (^{37}\text{Al}, ^{35}\text{Mg}\gamma)$ **2017Mo26**

$^{36}\text{Mg} \rightarrow 1n + ^{35}\text{Mg}$ from $J^\pi=0^+$ ^{36}Mg ground state.

$^{37}\text{Al} \rightarrow 1p1n + ^{35}\text{Mg}$ from $J^\pi=(5/2^+)$ ^{37}Al ground state.

2017Mo26: A secondary beam composed of ^{36}Mg and ^{37}Al was produced via the projectile fragmentation of a 345-MeV/nucleon ^{48}Ca primary beam impinging on a ^9Be target and selected by the BigRIPS separator at RIKEN. $E(^{36}\text{Mg})=235$ MeV/nucleon and $E(^{37}\text{Al})=246$ MeV/nucleon in front of the 2.54 g/cm² carbon secondary target. The reactions leading to ^{35}Mg from ^{36}Mg and ^{37}Al are likely 1n-knockout and 1p1n-removal reactions, respectively. The reaction residues were selected and identified by the Zero Degree spectrometer using the B ρ - ΔE -ToF method. The γ rays in coincidence with ^{35}Mg residues were detected using the DALI2 array of 186 NaI(Tl) crystals at 20°–150°. Measured $E_\gamma(>200\text{ keV})$, I_γ , $(^{35}\text{Mg})\gamma$ -coin, the inclusive one-neutron knockout cross section and exclusive γ -ray emission cross sections, and parallel momentum distributions of ^{35}Mg in coincidence with γ rays. Deduced levels, L-transfers, J, and π . Compared with shell-model calculations using the SDPF-M interaction in the sd shell with $\nu 1f_{7/2}$ and $\nu 2p_{3/2}$ orbits, and the SDPF-M interaction in a model space up to $\nu 2p_{1/2}$, and antisymmetrized molecular dynamics (AMD) model calculations using the Gogny D1S force.

 ^{35}Mg Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$L^\ddagger$	Comments
0			J^π : $3/2^-$ from shell-model calculations with the SDPF-M and SDPF-M+ $2p_{1/2}$ interactions; $3/2^+$ from AMD with the Gogny D1S force (2017Mo26).
0+x			$E(\text{level})$: <200 keV; the detection threshold in 2017Mo26 . The original placement in 2017Mo26 of all γ rays is to the ground state, which cannot be confirmed with existing data.
0+y?	($5/2^-$, $7/2^-$)		$E(\text{level})$: <200 keV; the detection threshold. 2017Mo26 suggested a low-lying level from the observed 42(1)% neutron knockout from the $l=3$ orbit in the inclusive parallel momentum distribution.
206+x 8			J^π : γ -ray intensity is too low to be assigned to the $1/2^-$ level at 141 keV from shell-model calculations with the SDPF-M+ $2p_{1/2}$ interaction (2017Mo26).
443+x 7	($3/2^+$, $5/2^+$)	(2)	
616+x 8	($1/2^-$, $3/2^-$)	(1)	
670+x 8			

[†] From E_γ data. **2017Mo26** suggested that all the four observed γ rays were emitted independently and fed either the ground state or a low-lying excited state below 200 keV. But the suggested experimental level scheme in FIG.4 of **2017Mo26** (with no γ rays shown) indicates all four γ rays feeding the ground state, which is not adopted by the evaluators since it cannot be confirmed with existing data.

[‡] **2017Mo26** deduced L by comparing the measured and eikonal-calculated parallel momentum distributions of ^{35}Mg residues. J^π options are deduced accordingly.

 $\gamma(^{35}\text{Mg})$

2017Mo26 states that no clear $\gamma\gamma$ coincidence was observed.

It is possible that the final levels of 206 γ , 443 γ , 616 γ , and 670 γ are not the same level from authors' comparisons with shell-model calculations with different interactions.

E_γ	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	Comments
206 ‡ 8	2 1	206+x		0+x	
443 ‡ 7	8 1	443+x	($3/2^+$, $5/2^+$)	0+x	2017Mo26 assigned this γ as the $3/2^+$ at 788 keV to $3/2^-$ g.s. transition in shell-model calculations with the SDPF-M+ $2p_{1/2}$ interaction (2017Mo26).
616 ‡ 8	7 1	616+x	($1/2^-$, $3/2^-$)	0+x	2017Mo26 assigned this γ as the $3/2^-$ at 664 keV to $1/2^-$ at 141 keV transition in shell-model calculations with the SDPF-M+ $2p_{1/2}$ interaction (2017Mo26).
670 ‡ 8	3 1	670+x		0+x	E_γ : From 2011Ga15 , as this γ is not resolved from the 616 γ in 2017Mo26 , but its presence is indicated in the fit of the spectrum. 2017Mo26 states that the origin of the 670 γ remains vague.

Continued on next page (footnotes at end of table)

C(³⁶Mg,³⁵Mgγ),(³⁷Al,³⁵Mgγ) 2017Mo26 (continued)

γ(³⁵Mg) (continued)

† γ-ray emission σ (mb).
‡ Placement of transition in the level scheme is uncertain.

